

A Randomized Controlled Trial of Varying Radiation Doses in the Treatment of Choroidal Melanoma

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Objective: To determine if a reduction in proton radiation dose from the standard dose of 70 cobalt gray equivalents (CGE) to 50 CGE would decrease radiation-induced complications, thereby improving visual prognosis, without compromising local tumor control for patients with uveal melanoma at high risk of these complications.

Design: Randomized, double-masked clinical trial.

Participants: A total of 188 patients with small or medium-sized choroidal melanomas (<15 mm in diameter and <5 mm in height) near the optic disc or macula (within 4 disc diameters of either structure).

Methods: Patients were treated with proton beam therapy at doses of either 50 CGE or 70 CGE between October 1989 and July 1994, and followed up biannually through April 1998. Outcomes included visual acuity, radiation complications, melanoma recurrence, and metastasis.

Results: Proportions of patients retaining visual acuity of

at least 20/200 were similar in the 2 dose groups at 5 years after radiation (approximately 55%). Similar numbers of patients in each group experienced tumor regrowth (2 patients at 50 CGE vs 3 patients at 70 CGE; $P>.99$) and metastasis (7 patients at 50 CGE vs 8 patients at 70 CGE; $P=.79$). Five-year rates of radiation maculopathy also were similar (for both groups, approximately 75% for tumors within 1 disc diameter and 40% for tumors >1 disc diameter from the macula). Rates of radiation papillopathy were nonsignificantly decreased in the 50-CGE treatment group when tumors were located 1 disc diameter or less from the optic disc ($P=.20$). Patients treated with the lower dose also experienced significantly less visual field loss.

Conclusions: This level of dose reduction did not result in a lesser degree of visual acuity loss. The lower-dose group did experience significantly less visual field loss. Local tumor recurrence and metastatic death rates were similar in both dose groups.

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MELANOMA of the uveal tract is the most common primary intraocular malignancy and the only potentially fatal intraocular tumor in the adult. Over the past few decades enucleation has been replaced by radiotherapy (RT) as the standard management of these tumors. Radiotherapy offers the advantage of preservation of the eye, and may conserve a useful level of vision, depending on the size of the tumor and its location with respect to the optic disc and macula. Two major types of radiotherapy are available: radioactive plaques that are sutured on the sclera over the area of the tumor,¹⁻⁴ and external beam irradiation using charged particles such as protons⁵⁻⁸ and helium ions.⁹ These modalities, although effective in achieving local tumor control, may be associated with high rates of morbidity, particularly when the tumor is large or is located near the optic disc or fovea.¹⁰

Adverse effects of proton beam therapy include radiation retinopathy and papil-

lopathy,¹¹ loss of vision,^{10,12} cataract,^{13,14} and, infrequently, enucleation.¹⁵ Factors positively associated with an increased risk of vision loss and radiation vasculopathy, in particular, include tumor height and proximity to the optic disc or fovea.

We conducted a randomized, double-masked clinical trial to determine if a reduction in the radiation dose from the standard 70 cobalt gray equivalent (CGE) to 50 CGE (CGE=proton dose in gray × radiobiological effectiveness=1.1) would reduce radiation-induced complications without compromising local tumor control. The study was limited to small to moderate-sized melanomas located near the fovea or optic disc. We thought a priori that this subgroup, at high risk of radiation-related complications, would gain the most advantage from a reduction in dose, without detriment to local control.

RESULTS

Table 1 displays baseline demographic and clinical characteristics of the patients, by

PATIENTS AND METHODS

Patients with melanoma of the choroid and/or ciliary body located within 4 disc diameters (DD) (approximately 6 mm) of the optic disc, the macula, or both structures were included in the study. In addition, because biannual examinations were planned, eligible patients had to reside in the United States or Canada. Exclusion criteria included presence of metastatic disease at the time of treatment, and prior treatment for the intraocular tumor. In addition, large tumors, defined as those 15 mm or greater in diameter or 5 mm or greater in height, were excluded, because such tumors are associated with poor visual prognosis.¹²

All patients were recruited from the Ocular Oncology Clinic of the Retina Service at the Massachusetts Eye and Ear Infirmary, Boston. During 5 years of recruitment (October 1989 through July 1994), 188 of 252 eligible patients were enrolled in the trial (participation rate, 74.6%). This sample size yielded a power of 80% to detect a 50% reduction in the rate of visual loss (to worse than 20/200), from 40% to 20%, and an increase in the rate of local recurrence from 3% to 12% in the lower-dose group by 3 years after irradiation ($P = .05$).

A complete description of the protocol for proton beam irradiation of intraocular melanomas at the Harvard Cyclotron Laboratory has appeared elsewhere.⁶ All aspects of surgery for tumor localization were identical in the 2 treatment groups. All patients received proton beam therapy in 5 fractions, generally delivered over a 7-day period, in collaboration with the Department of Radiation Oncology at Massachusetts General Hospital and the Harvard Cyclotron Laboratory. However, for those in the lower-dose group, each fraction was reduced by a factor of 0.714 relative to the standard dose.

Randomization was performed using a stratified random permuted block design.¹⁶ Stratification by tumor height (<3 mm vs ≥ 3 mm) and posterior location of the tumor (abutting the disc or macula, >0 -1 DD, >1 -2 DD, or >2 -4 DD from both structures), characteristics highly predictive of visual outcome, was performed to ensure that the treatment arms would be well balanced. Within each stratum, patients were assigned at random, in blocks of 4, to receive either 50 CGE or 70 CGE. Dose assignments were determined by research staff, who notified the attending physician of the names and dose assignments of each new participant. Periodic audits were conducted to ensure that patients had been treated with the intended dose.

Both the treating physician and patient were masked to the dose assignment. The treatment assignment was disclosed only in the event of tumor regrowth, patient death, or, infrequently, upon request of the patient. To ensure that the physician remained masked to the dose assignment, all documents containing any reference to dose assignment were maintained in separate files that were inaccessible to the treating physicians.

The study was reviewed and approved by the Human Subjects Committees of the Massachusetts Eye and Ear Infirmary and the Massachusetts General Hospital. Study investigators met twice yearly to review recruitment and

compliance statistics as well as to identify any safety concerns relating to lowering the dose (eg, a marked increase in local recurrence or metastasis).

DATA COLLECTION

Information collected at baseline on these patients has been described in detail previously.¹⁷ In brief, demographic information including age at treatment and sex was recorded for all patients. Tumor size (largest tumor diameter and height) was estimated based on indirect ophthalmoscopy, transillumination during surgical localization, and ultrasonography. Tumor location in relation to the optic disc, macula, equator, and ora serrata also was determined. Treatment was planned with a standard program in all patients. Dose to structures (optic disc, macula, lens, retina, ciliary body) was estimated based on isodose curves generated from treatment plans.¹⁸

Patients were followed up biannually through April 1998. Each ophthalmological examination included measurement of visual acuity (using a standard nursing protocol), tumor height (using A-scan ultrasonography), and intraocular pressure. Visual acuity was designated as the line on an ETDRS (Early Treatment Diabetic Retinopathy Study) chart in which the patient could read a minimum of 3 letters; we also recorded the total number of letters read (0-110). The examiner for all measures was masked to the patient's treatment assignment. The examining physician recorded the presence and severity of lens opacity and other radiation-associated complications, and whether there was evidence of tumor regrowth, based on a combination of ultrasonography, ophthalmoscopy, and sequential fundus photographs of tumors and nearby landmarks. Fundus photography, fluorescein angiography, and visual field testing were performed on an annual basis. All photographs were obtained using the ETDRS protocol.¹⁹ Assessment of the central visual field (30°) in patients with a minimum visual acuity of 20/200 was completed using the G1 program of the Octopus perimeter (Interzeag, Inc, Northborough, Mass). The majority of examinations were conducted at the Massachusetts Eye and Ear Infirmary. When necessary, biannual examinations were scheduled with the referring ophthalmologist, who was asked to forward results to us.

DATA ANALYSIS

Differences in vision loss, radiation complications, and visual field defects between the 2 dose groups were calculated at annual time points up to 5 years after irradiation. Vision loss was evaluated in terms of the proportion retaining a minimum acuity of 20/200 (among patients with a baseline acuity of at least 20/200), and number of letters lost relative to baseline vision. Percentage of patients retaining visual acuity of at least 20/200 and rates of onset of radiation papillopathy and maculopathy were estimated using Kaplan-Meier formulations²⁰; equality of rates was tested using the log-rank test.²¹ Visual field loss in the central 30° was defined as the increase in defect (in decibels), averaged across test points, from the baseline examination to annual time points after irradiation. Differences in the average mean increase in mean defect between the 2 dose groups were assessed using the t test.

dose assignment. Treatment arms were generally well balanced. The conventional dose (70 CGE) arm had a significantly higher proportion of men ($P = .04$), but the groups were similar in age and residency. Statistically significant

though generally modest differences were observed in factors predictive of tumor-related survival: those assigned to the lower dose had slightly larger (in basal diameter) and more anterior tumors. Patients were recruited for the trial

Table 1. Baseline and Treatment Characteristics of Participants According to Assigned Dose*

	50-CGE Group (n = 94)	70-CGE Group (n = 94)	P
Demographic characteristics			
Age	62 (19-86)	57 (27-84)	.37
Male	44 (47)	59 (63)	.04
Residence in New England	46 (49)	51 (54)	.56
Predictors of metastasis			
Largest tumor diameter, mm	11.0 (7.0-16.0)	10.0 (7.0-17.0)	.03
Perpendicular diameter, mm	8.0 (4.0-15.0)	7.0 (4.5-13.0)	.09
Anterior tumor margin			
Anterior to equator	16 (17)	6 (6)	.04
Extrascleral extension	1 (1)	1 (1)	1.00
Predictors of visual outcome			
Tumor height, mm	3.0 (1.2-6.3)	3.0 (1.0-5.5)	.89
Distance to disc, DD	1.4 (0-6.0)	1.3 (0-5.5)	.67
Distance to macula, DD	1.0 (0-7.0)	0.9 (0-6.0)	.50
Macular detachment	13 (14)	15 (16)	.69
Visual acuity	20/32 (16-800)	20/32 (16-HM)	.54
Dose to structures			
Optic disc, CGE	31.2 (0-50)	42.0 (0-70)	.02
Macula, CGE	45.0 (0-50)	67.9 (0-70)	.00
Percentage exposed to >50% of dose			
Lens	0.0 (0-28)	0.0 (0-38)	.67
Retina	40.0 (8-67)	41.5 (12-68)	.71
Ciliary body	24.0 (0-40)	24.0 (0-51)	.80

*Data are given as median (range) or number (percentage) of patients. CGE indicates cobalt gray equivalent; DD, disc diameter; and HM, hand motions.

based on tumor dimensions obtained during the clinical examination. These dimensions were then confirmed, after enrollment, during surgery for tantalum ring placement. As a result, tumor size (height or diameter) slightly exceeded the limits for eligibility in 4 patients. Better balance was achieved on factors associated with visual outcome, including tumor height, distance of tumor from optic disc or macula, and baseline visual acuity. Similar proportions of the lens, retina, and ciliary body were included in the radiation field. As expected, doses to the optic disc ($P=.02$) and macula ($P=.00$) were significantly lower in the 50-CGE group.

The majority of patients completed scheduled bi-annual examinations, and good compliance was achieved in both treatment arms (**Table 2**). The proportion of examinations completed at Massachusetts Eye and Ear Infirmary declined with time following irradiation; by 5 years after treatment, 33% and 38% of all examinations were performed by the referring ophthalmologist in the 50-CGE and 70-CGE groups, respectively.

Table 3 presents visual acuity results based on the ETDRS chart examinations. Visual outcome was similar regardless of radiation dose. At 5 years after treatment, the median visual acuity was 20/160 in the 50-CGE treatment arm and 20/100 in the standard dose arm ($P=.91$, Wilcoxon rank sum test). Approximately 55% of patients in both groups retained a visual acuity of 20/200 or better in the treated eye (**Figure 1**). When evaluated in terms of decline in letters read relative to the baseline examination, the median loss of letters was 13 and 27, in the low- and standard-dose groups, respectively ($P=.61$, Wilcoxon rank sum test).

Table 2. Compliance With Follow-up Examinations*

		50-CGE Group	70-CGE Group	P
Baseline	No. randomized	94	94	
Months after irradiation				
12	No. eligible†	94	93	
	Examined	93 (99)	90 (97)	.99
	MEEI	85 (91)	82 (91)	
	Elsewhere	8 (9)	8 (9)	
24	No. eligible†	90	89	
	Examined	84 (93)	86 (97)	.40
	MEEI	73 (87)	70 (81)	
	Elsewhere	11 (13)	16 (19)	
36	No. eligible†	84	85	
	Examined	81 (96)	80 (94)	.71
	MEEI	64 (79)	61 (76)	
	Elsewhere	17 (21)	19 (24)	
48	No. eligible†	81	82	
	Examined	75 (93)	67 (82)	.57
	MEEI	57 (76)	48 (72)	
	Elsewhere	18 (24)	19 (28)	
60	No. eligible†	80	81	
	Examined	61 (76)	53 (65)	.69
	MEEI	41 (67)	33 (62)	
	Elsewhere	20 (33)	20 (38)	

*CGE indicates cobalt gray equivalent; MEEI, Massachusetts Eye and Ear Infirmary. Data are given as No. (%) unless otherwise indicated.

†Number of patients surviving without enucleation who were treated a minimum of 12, 24, 36, 48, or 60 months prior to study closeout (April 30, 1998). One patient in the high-dose group died before the scheduled 12-month follow-up visit.

Visual field results, measured in terms of average increase in the threshold for light perception (in decibels) across test points in the G1 program of the Octopus perimeter, suggested some benefit of lowering the radiation dose (**Figure 2**). The low-dose group experienced less deterioration in visual field. The difference in the mean increase in defect between the dose groups was less than 1 decibel at 1 year after irradiation. With additional follow-up, the mean increase in defect was 2.3 (95% confidence interval, 1.1-8.2) to 3.2 (95% confidence interval, 0.94-31.4) times higher in the standard dose group than in the lower-dose group.

By study design, patients were at high risk of radiation papillopathy and maculopathy given the proximity of their tumors to the optic disc and macula. To assess differences in rates of these complications, we calculated stratum-specific (≤ 1 DD vs > 1 DD from the macula or optic disc) Kaplan-Meier estimates of the cumulative percentage free of these complications according to year after treatment (**Figure 3** and **Figure 4**). For clinically assessed maculopathy (**Figure 3**), incidence rates depended on distance of the tumor margin from the macula, but not on the radiation dose. In both arms, the 5-year rate of maculopathy was approximately 75% when the tumor extended within 1 DD (**Figure 3**, top), and approximately 40% when the tumor was at greater distances (**Figure 3**, bottom) from the macula. For papillopathy (**Figure 4**), we observed some improvement by lowering the dose among tumors located within 1 DD of

Table 3. Visual Acuity Based on ETDRS Chart Examination*

		50-CGE Group	70-CGE Group	P
Baseline	Visual acuity, median	20/32	20/32	.26
	Letters read, median	95	95	.25
Months after irradiation				
12	Visual acuity, median (Q1-Q3)	20/40 (20/25-20/80)	20/31 (20/20-20/71.5)	.14
	Letters read, median (Q1-Q3)	90 (73-100)	97 (76-102)	.13
	Letters decline,† median (Q1-Q3)	4 (-4 to 11)	2 (-4 to 9)	.39
	≥20/200, No. (%)	83 (92)	81 (96)	
24	Visual acuity, median (Q1-Q3)	20/50 (20/25-20/400)	20/50 (20/25-20/200)	.92
	Letters read, median (Q1-Q3)	84 (45-100)	84 (55-99)	.92
	Letters decline, median (Q1-Q3)	6 (0-40)	7 (0-32)	.61
	≥20/200, No. (%)	65 (79)	68 (86)	
36	Visual acuity, median (Q1-Q3)	20/56 (20/25-20/800)	20/80 (20/25-20/800)	.81
	Letters read, median (Q1-Q3)	77 (26-99)	72 (30-100)	.90
	Letters decline, median (Q1-Q3)	12 (0-43)	20 (1-51)	.33
	≥20/200, No. (%)	52 (68)	49 (69)	
48	Visual acuity, median (Q1-Q3)	20/100 (20/25-20/500)	20/100 (20/25-20/800)	.82
	Letters read, median (Q1-Q3)	69 (27-97)	71 (28-99)	.55
	Letters decline, median (Q1-Q3)	18 (1-50)	17 (2-60)	.76
	≥20/200, No. (%)	44 (61)	35 (61)	
60	Visual acuity, median (Q1-Q3)	20/160 (20/25-20/900)	20/100 (20/25-20/900)	.91
	Letters read, median (Q1-Q3)	60 (25-98)	62 (25-95)	.86
	Letters decline, median (Q1-Q3)	13 (0-56)	27 (2-55)	.61
	≥20/200,‡ No. (%)	33 (56)	24 (54)	.82

*CGE indicates cobalt gray equivalent; Q1-Q3, the interquartile range describing the bounds of the 25th and 75th percentiles.

†Letters decline = baseline letters read – current letters read.

‡Kaplan-Meier estimate of cumulative proportion retaining a minimum visual acuity of 20/200 at 12, 24, 36, 48, and 60 months after irradiation. P value is based on log-rank test.

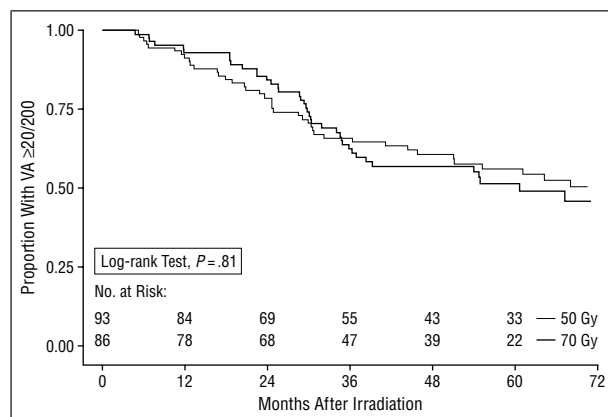


Figure 1. Kaplan-Meier curves for visual acuity (VA) retention of 20/200 or better at time points after irradiation.

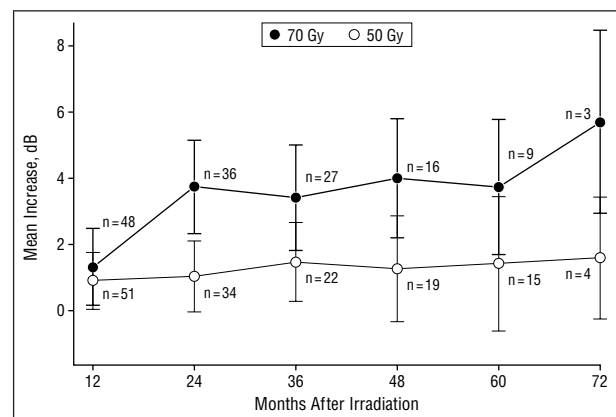


Figure 2. Mean (and 95% confidence intervals) increase from baseline in average defect across test points in the G1 program of the Octopus perimeter, at time points after irradiation, among patients with a visual acuity of 20/200 or better.

the optic disc (Figure 4, top): at 5 years after treatment, the rate was 58% in the low-dose arm and 77% in the higher-dose arm ($P=.20$). Few patients (approximately 7% in each dose group) with less proximal tumors developed signs of optic disc damage (Figure 4, bottom).

Rates of other radiation complications are shown in **Table 4** by dose assignment. No statistically significant differences were observed. The slight excess of posterior subcapsular opacities, rubeosis, and enucleations in the lower-dose group may have been observed because these tumors were, by chance, slightly larger and more anterior.

There were no differences by dose in number of local recurrences ($P>.99$) or metastatic deaths ($P=.79$). Approximately 10% of patients in each arm experienced

local recurrence or metastatic spread of the tumor (**Table 5**), which is consistent with data based on the larger cohort of patients with similar tumors treated at the standard dose (70 CGE).

Regression in tumor height was observed in the majority of patients by 4 years after treatment, with steadily more tumors showing decline at each annual examination (**Table 6**). The data suggest more rapid shrinkage in the tumors treated with high-dose radiation, although differences by dose were not statistically significant. By 5 years after treatment, measureable regression (minimum of 0.5 mm) was observed in 69% and 76% of tumors in the low- and higher-dose groups, respectively ($P=.33$).

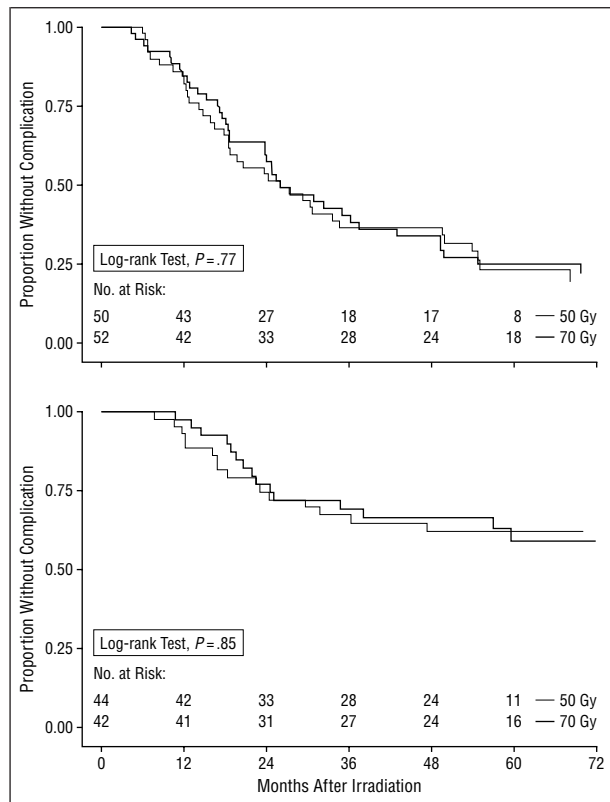


Figure 3. Kaplan-Meier curves for the cumulative proportion of patients without radiation maculopathy at time points after irradiation, according to distance of the tumor from the macula.

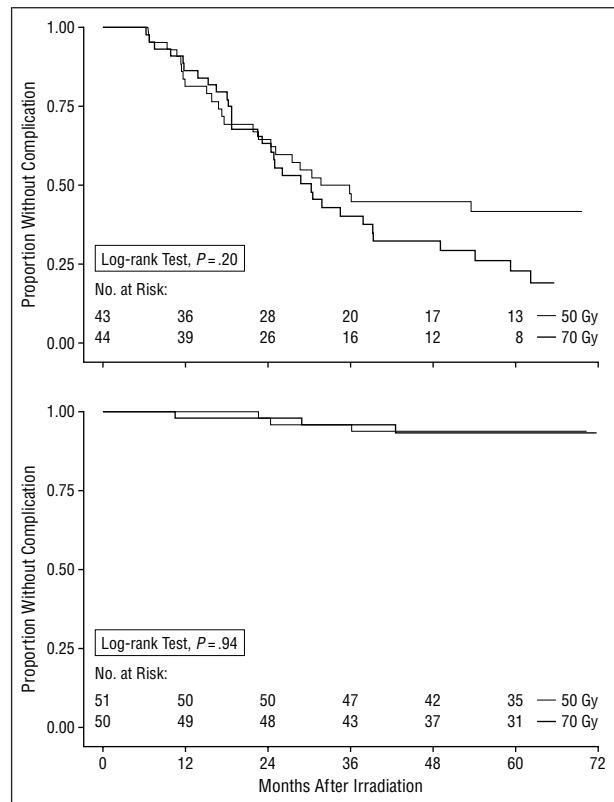


Figure 4. Kaplan-Meier curves for the cumulative proportion of patients without radiation papillopathy or atrophy at time points after irradiation, according to distance of the tumor from the optic disc.

Table 4. Other Complications and Enucleation*

	No. (%) of Patients	
	50-CGE Group	70-CGE Group
Vitreous hemorrhage	14 (15)	12 (13)
Subretinal exudation in macula	10 (11)	8 (8)
Posterior subcapsular opacity	13 (14)	7 (7)
Rubeosis/neovascular glaucoma	9 (10)	7 (7)
Radiation keratopathy	2 (2)	4 (4)
Uveitis	0	1 (1)
Enucleation	4 (4)	5 (5)

*CGE indicates cobalt gray equivalent.

Table 5. Local and Systemic Tumor Recurrences Within 5 Years After Irradiation*

	No. (%) of Patients	
	50-CGE Group	70-CGE Group
Tumor regrowth		
Marginal recurrence	0	1
Uncontrolled	1	1
Ring melanoma	1	0
Undocumented	0	1
Total†	2 (2)	3 (3)
Metastasis‡	7 (7)	8 (8)

*CGE indicates cobalt gray equivalent.

†P > .99.

‡P = .79.

COMMENT

Based on the findings of this study, it does not appear that a radiation dose reduction of 30%, from 70 CGE to 50 CGE, significantly improves functional outcome in patients with small to moderate-sized tumors near the optic disc or macula in this treatment setting. However, modest improvements are suggested for 2 outcomes: visual fields and radiation papillopathy. Results with respect to the optic nerve support our previous observations that radiation papillopathy or atrophy is largely confined to the relatively small subgroup in whom the optic disc receives the full irradiation dose.

Level of radiation exposure received by the macula was reduced by an average of 28 CGE in the low-dose arm. However, we were unable to demonstrate any ac-

companying improvement in maculopathy rates with this dose reduction. One possible explanation for this outcome is that 50 CGE exceeds the threshold for radiation effects in the macula. We have shown previously that the probability of developing maculopathy rises linearly with dose to the macula, plateauing at approximately 40 CGE.¹¹ In this trial, more than half of patients in the 50-CGE group received a dose to the macula of 40 CGE or higher. Given this fact, the experimental dose may have been too high to achieve an appreciable decline in rates of maculopathy. Subtle differences in the degree of radiation maculopathy in the low-dose arm may be demonstrable in fundus photographs and fluorescein angiograms, when those results are available.

Table 6. Tumor Height Regression in Patients Examined at Massachusetts Eye and Ear Infirmary*

		50-CGE Group	70-CGE Group	P
Baseline	Mean height, mm	3.05	3.04	.91
Months after irradiation				
12	No. of patients	85	82	...
	Mean height, mm	2.40	2.22	.25
	Mean decline, mm	0.65	0.80	.24
	% Regressed†	30	21	.18
24	No. of patients	73	70	...
	Mean height, mm	2.04	1.79	.10
	Mean decline, mm	0.91	1.07	.27
	% Regressed†	38	37	.88
36	No. of patients	64	61	...
	Mean height, mm	1.68	1.69	.98
	Mean decline, mm	1.23	1.16	.58
	% Regressed†	46	47	.88
48	No. of patients	57	48	...
	Mean height, mm	1.67	1.57	.58
	Mean decline, mm	1.29	1.35	.75
	% Regressed†	50	59	.24
60	No. of patients	41	33	...
	Mean height, mm	1.59	1.54	.85
	Mean decline, mm	1.47	1.30	.50
	% Regressed†	69	76	.33

*CGE indicates cobalt gray equivalent.

†Percentage with regression of at least 0.5 mm based on ultrasound examination.

Based on 5-year follow-up, our data suggest that there are no deleterious effects of lowering the dose; by 5 years after treatment, similar rates of local tumor recurrence and metastatic spread were observed in both dose groups. These findings were not surprising given that our estimate of tumor control probability at 50 CGE was approximately 0.93 to 0.95 (based on the observed tumor control probability of 0.99 at 70 CGE, which included recurrences within the treated volume). This estimate for tumor control probability is in accord with the estimates by Bentzen et al²² for malignant melanoma of the skin and lymph nodes. Their radiation dose estimates for achieving control in 80% of tumors (TCD80 values) were 22 Gy and 28 Gy, delivered in 9-Gy fractions for 10- and 20-mm lesions, respectively. Thus, the lower dose in our trial (50 CGE in 5 fractions) is approximately double their estimated TCD80.

It is possible that with continued follow-up we will observe a significant late beneficial effect in the 50-CGE group overall, or in a specific subgroup. It is also possible that we missed true differences by dose, particularly in underpowered subgroup analyses. As noted, 50 CGE may exceed the physiologic threshold for untoward biologic effects in the eye below which the relationship may be more linear. However, on the basis of present results, it appears that dose reduction to 50 CGE may not produce the desired result of improving functional outcome in the subset of eyes in which high radiation exposure to the fovea and optic disc is unavoidable. Although further reductions in dose could be considered, it should be done with great caution as failure to achieve local control may increase the risk of tumor recurrence and early death from metastasis.²³ An alternative approach, which we plan to in-

vestigate, is to increase the number of fractions and lower the dose per fraction, which, by lowering the effective dose to normal tissues, may optimize functional outcome in these patients.

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